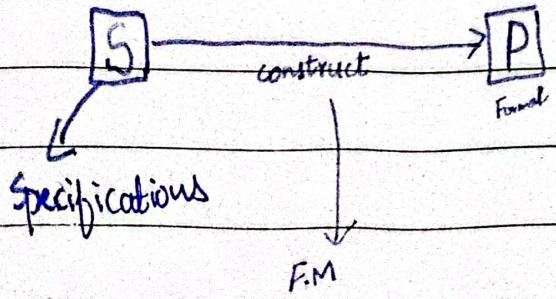
FORMAL METHODS IN SOFTWARE ENGINEERING
BSE: 8th SEMESTER

FORMAL METHODS

* To make the system works better by using mathematical methods.
* Validating the software using mathematical operations.
* Assignments, Quizzes, Projects.
* **Why Formal Methods:**

- Production
- staging



* Operations / services } Confusion, less clarity → Informal work
* FCL (Formal Classification Language)
* Three Levels:
  L0 - Formal specify L1 - Formal verification
  L2 - Formal transforming.

29/1/26

⊙ Reliability and Dependability:

→ Reliability is subset of dependability

→ Faulty behavior less (Reliability)

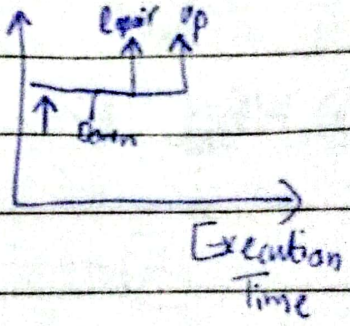
⊕ KEY CONCEPTS:

Reliability
- Fault ····> system Problem
- Failure
- Time

Real-world time
CPU Time (Execution Time)

Models:
1) log
2) Basic

(System (start of operations))



⊛ Stages: (3)

→ MTTF
→ MTTR
→ MTBF

	Up	Down
MTTF	25 sec	15 sec
	15 sec	10 sec
	10 sec	
	⋮	⋮
MTTR	15 sec	

MTBF

⊛ $\lambda(t) = \frac{d(U(t))}{t}$

⊛ Applications of Software Reliability:

(i) Comparison of software technologies.
(ii) Measure progress of system testing.
(iii) Controlling the system in operation
(iv) Insights of detailed process.

⊛ Decrement Processes:

(2)

- Decrement Process 1
  - ↳ Basic Model (linear graph)
- Decrement Process 2
  - ↳ Log Model (curve graph)

⊛ Dependable System:

- Users trust upon System
  (Trust ↑ Dependency ↑)
- Dependability:
  (i) No failures (ii) Expected output

⊙ Causes of failure:

- Software failure
- Hardware failure
- Operational failure

⊙ Dependability ↑ Cost ↑

3/2/26

- Formal Methods — Overview, Benefits, Applications
- Usability & Criticism of Formal Methods.

⊙ Benefits and Applications:

→ It draws the upper and inner characterness of software.
→ It deals with correctness not with the quality.
→ Reducing the number of errors
→ Cost effective, concise, validate (mathematical model)

① Requirements definition → Requirements
Modeling ← specifications
② Specification and Design
③ Validation is required to achieve from Formal Methods.
④ Specifications Technique:
(i) Algebraic Approach
(ii) Model-based Approach.

⊙ Logic Symbols:

★ (for all) $x, y, z : x > y \wedge y > z \rightarrow x > z$

E.g:
```
var x, v;
func add (x, v):
    return x+v;
```

: Pre-condition

1) Input $(v, x) > 0$

↓

Operation → $(v + x = z > 0)$

$\forall v, x > 0 \; (v + x) \longrightarrow > 0$

10/2/26

⊙ Usability of Formal Methods

→ Tools are used to sleek the formal methods.

→ Usability is related with reliability and it is achieved using different mathematical expressions.

⊙ Limitations:

→ Poor quality of integration tools are available.

→ Special trained person is required for formal methods in software.

→ Microservices of software production will take lot of time.

⊛ Technology Transfer:

→ Idea or innovation developed into practical thing.

→ Effective Scalable Method transfer the gap between the Researchers (PhD) and developers.

→ There will be gap between the transfer i.e the talk and perception of the words.

17/2/26

SET THEORY

⊛ Introduction:

Use → Computer science, Database, (Union, Inter section)

Operating System, Data structure, Cybersecurity
(Scheduling) (Hashing)

→ Distinct unordered collection of elements (Def)

→ Relations: Union, Intersection

→ operations:

⊙ Notations:

(i) Set Builder

(ii) Tabular.

→ Unordered: {a, b, c}, {c, b, a}

⊛ Tabular:

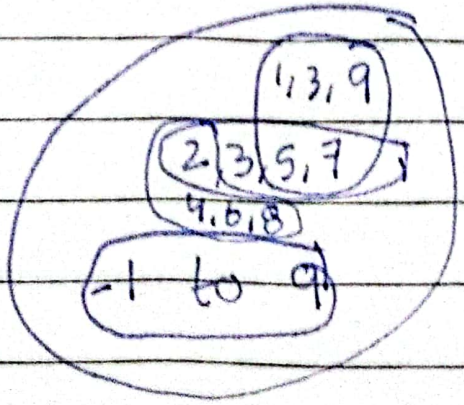
→ Z^+ (Positive Integers)

: $\{x \mid x > 0 ; Z^+\}$

→ {1, 2, 3, 4} : Cardinality

: $\{x \mid x > 0 \wedge x < 5\}$

→ Integers −1 to 9:



$\forall S, T : S = T \overset{\text{(only if)}}{\longleftrightarrow} \{\forall x : x \in S \longleftrightarrow x \in T\}$

⊛ Singleton (Single element set).

(*) $S \to \{1,2,3,4\}$

$T \to \{1,2\}$

$T \subset S$

Proper Improper

(*) $S \subseteq T \wedge T \not\subseteq S$ (Proper) (one more element)

$S \subseteq T \wedge T \subseteq S$ (Improper)

(*) Power of set = $P(S) = |P(S)| = 2^5$

19/2/26

(*) **UNION:**

$\to \forall A, B: A \cup B = \{x \mid x \in A \vee x \in B\}$

$A = \{1,2,3\}$

$B = \{1,2,3,4,5\}$

$A \cup B = \{1,2,3,4,5\}$

(*) Intersection, Functions, Relations, Types, Applications.

(*) **CARTESIAN PRODUCT:**

$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$

(a,b) → ordered pairs

A = {1, 2, 3}
B = {False, True}

A x B = {(1, F), (1, T), (2, F), (2, T), ..., }

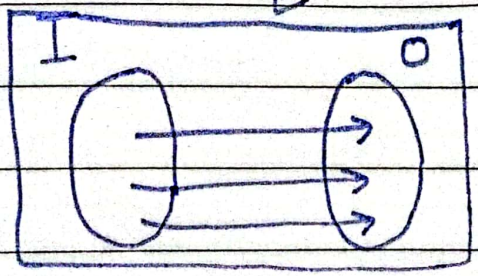
R ⊆ A x B

(subset)

* DOMAIN (First Element), RANGE (Second element)

* FUNCTION TYPES:

→ Types → Injective (one to one)
→ Bijective (Every input has some output)
→ Surjective (Injective + Bijective)



* Relation Composition:

→ S : A x B r : B x C

S : A x C

3/3/26

→ Logic History / Evaluation :

- Two millineals
- Aristotle
  - Philosophy
  - Using all/some/none
  - Premise 1
    Premise 2
    Conclusion
    ↓
    Deductive Reasoning
  - P1: All students are humans
  - P2: All humans are mortal
- Stoic
  - Propositional variables
  - Modus Tollis
    - $p \rightarrow q \equiv q$
  - Modus Ponin
    - $p \rightarrow q \Rightarrow \neg p$
- Boole
  - Boolean Algebra
  - (0, 1) Binary
  - DLD
- Frege
  - First order predicate logic
    $\Rightarrow \forall, \exists$
  - Variables, Functions

→ Logic:

- Truths are unpredictable
- Complexity
  ↓
  Model
  ↓
  Limitations of Logic

: Transport Industry

→ Propositional Logic:

- → Any Declarative statements that results in T/F
- → Any connectives:
  - $\vee$ OR
  - $\wedge$ AND
  - $\neg$ NOT
  - $\rightarrow$ Implies
  - $\leftrightarrow$ If and only if

→ Tautology and Contradiction

5/3/26

⊛ Proof and Properties in Propositional Logic:

→ Properties → Proofs Methods

→ Laws

→ Tautology

$P \vee \neg P$

P V −P → Tautology

P ∧ −P → Contradiction

It rains — It rains

: T or

→ Contradiction / Negative / False

$P \wedge \neg P$

It is raining — It is not raining.

⊛ Logical Equivalence:

- Truth values
- $P \leftrightarrow Q$ (Tautology)

e.g: $\neg\neg P \equiv P$

$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$

P	Q	$P \rightarrow Q$	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$
T	T	T	F	F	T
T	F	T	T	F	T
F	T	F	F	T	F
F	F	T	T	T	T

⊛ Laws:

→ Idempotent:

⊙ $P \wedge P \equiv P$, $P \vee P \equiv P$

→ Identity:

⊙ $P \wedge T \equiv P$, $P \vee F \equiv P$

(T: True, F: False)

→ Domination:

⊙ $P \vee T \equiv T$, $P \wedge F \equiv F$

→ Commutative:

⊙ $P \wedge Q \equiv Q \wedge P$

$P \vee Q \equiv Q \vee P$

→ Associative:

⊙ $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$

⊙ $(P \vee Q) \vee R \equiv P \vee (Q \vee R)$

→ Distributive Law:

⊙ $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

⊙ $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

⊙ E.g: It is raining & (cold or windy)

↓

(Rain & cold) OR (Rain & Windy)

10/3/26 (online)

→ **De Morgan's Laws:**

⊙ $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

→ Modus Ponens:

$P \rightarrow Q$

P $\therefore Q$

→ Modus Tollens.

$P \rightarrow Q$ $\therefore \neg P$

$\neg Q$

→ Hypothetical syllogism:

$P \to Q$

$Q \to R \quad \therefore P \to R$

→ Disjunctive syllogism:

$P \vee Q$

$\neg P \quad \therefore Q$

12/3/26

⊛ Propositional function:

→ x is greater than 3

subject predicate

→ It will give us true or false value.

→ Quantifiers make it proposition for a range of values

e.g. $\exists x \sqrt{x} < 4$

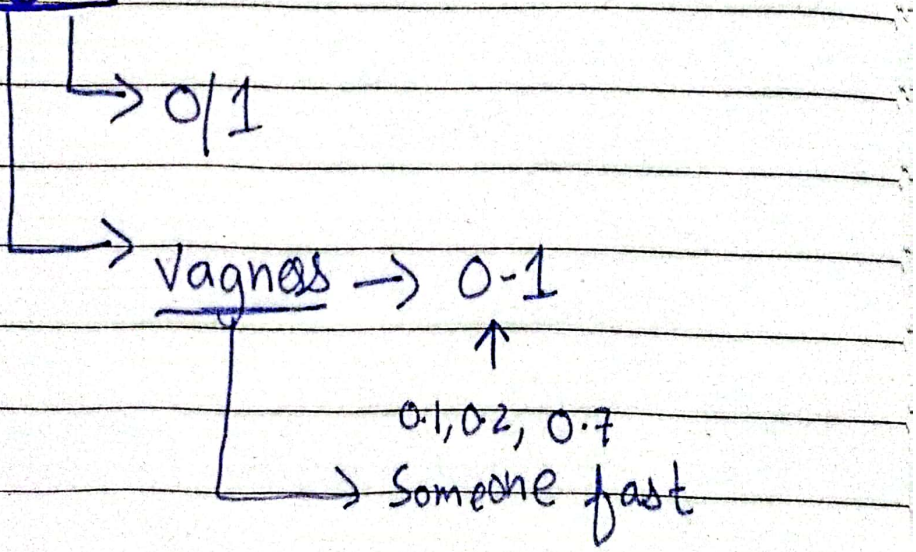
▷ UoD = {10, 12, 14, 16, 18, 20}

: only true for {10, 12, 14}

3/4/26

→ Semantic Tableaux
  - Rules
→ Applications/ Summary
→ Advance Logic
  - Fuzzy Logic.
  - Temporal Logic
  - Applications

① Logic:



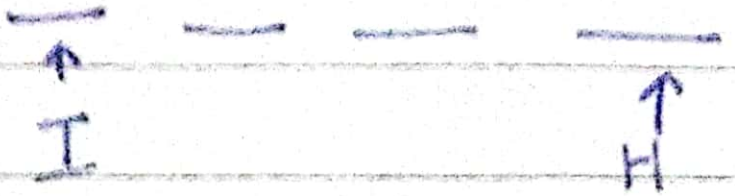
* Modd → Human Reasoning
  to

Vagness
  Complexity
    UC → R, I, F
      → ∀, ∃
  → tall, warm, expensive

⊙ Fozzy Logic Apps:

- Control systems → AC, Washing Machine
- AI → DT, RF
- Image Processing → Edge Detection, Image Enforcement
- Natural Language
  - ↳ LLMS, NLP
    - ↓
    - Transformers
    - ↓
    - Encoder / Decoder
- AI
  - ⊕ Supervised
  - ⊛ Unsupervised
- DNNs
  - Deep Neural Network (Hidden Layer)
- ANNs
  - Artificial Neural Network
  - ↳ Bias / Output Reasoning
- Tall (Quantification)
  - ↳ ↑↓ Highly Tall → (180cm)
  - Fairly Height → (100cm)
- Temporal Logic (Linked with Time) (Over Time)

e.g ↑ Request



- LTL
  Linear Temporal Logic
- CTL
  Computation Temporal Logic
  ↓
  Path putation
  Multiple formats
  Branching